
INVESTMENT BRIEF

Lupine Science

Computational Materials Intelligence. Three Revenue Streams. One Platform.

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June 2026

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Executive Summary

Lupine Science builds software that reads the geometric fingerprint of prediction errors in atomic-scale simulations. Our core asset is **atlas-distill** — a Rust engine that compresses months of materials benchmarking into hours of actionable intelligence.

We own the largest systematic error-geometry dataset in the field: **559 interatomic potentials × 15 elements × 1,677 data points**. We have published the first proof that these errors collapse onto low-dimensional **"hyper-ribbon" manifolds** across both classical and machine-learning potentials. This is not a hypothesis. It is a demonstrated mathematical structure with direct commercial applications.

Three revenue streams:

1. **Software licensing + consulting** — Yearly licenses for atlas-distill plus implementation consulting for national labs and materials companies
2. **IP licensing for discovered materials** — Patent pipeline for computationally discovered compositions, licensed to manufacturers
3. **Investment trading strategies** — Quantitative signals derived from open atomic-simulation research data, deployed via family office and institutional vehicles

Materials informatics is a **\$2.4B market growing at 13.4% CAGR**. The convergence of ML interatomic potentials, open benchmark databases, and sloppy-model theory creates a narrow window for a platform company that owns the error-analysis layer. Lupine Science is building that platform.

Seed ask: \$8M. 24-month runway to product-market fit across all three revenue streams, a portfolio of composition patents, and a live trading track record sufficient for Series A.

Revenue Stream 1: Software License + Consulting

Product. atlas-distill v0.1.0 is a Rust-based engine that performs five operations no competing platform offers in one toolchain:

- **Principal component analysis** of prediction-error manifolds
- **Hyper-ribbon detection** — automatic classification of low-dimensional error structure
- **Random-effects meta-analysis** — pooled inference across heterogeneous benchmarks
- **Simpson's paradox / ecological fallacy detection** — the only tool in the field with this
- **Bootstrap uncertainty quantification** — auditable confidence intervals for regulated applications

Pricing Model. Tiered yearly licensing by compute volume and organization type, plus implementation consulting for deployment, custom pipelines, and team training. Pricing set based on value delivered — six months of manual benchmarking compressed to 48 hours.

Target Customers. National labs (NIST, Argonne, Oak Ridge, Sandia), semiconductor companies (Intel, TSMC, Samsung), defense/aerospace (Boeing, Lockheed Martin), and materials startups (battery companies, catalyst developers).

Differentiation. The only platform built on sloppy-model theory. The only platform with formal Simpson's-paradox detection. Open-source core (MIT) drives adoption; enterprise features drive revenue. IMMI publication establishes scientific credibility.

Traction. GitHub repository public since 2025. Inbound inquiries from OpenKIM consortium members. One enterprise pilot in active discussion. No paid marketing.

Revenue Stream 2: IP Licensing for Discovered Materials

Concept. Hyper-ribbon error geometry is a **discovery signal**. Elements with high participation ratio ($PR > 1.5$) are "underparameterized" — many potentials exist, but none capture the full physics. These elements sit at the frontier of model failure, which means they also sit at the frontier of material discovery.

Process Pipeline.

1. **Screen** — Hyper-ribbon analysis across OpenKIM and Materials Project to identify high-PR elements (Fe, V, Cr in BCC; Al, Pd in FCC)
2. **Predict** — Deploy MACE-MP-0, CHGNet, Orb-v3 to predict properties of novel compositions
3. **Validate** — DFT validation on promising candidates (VASP, Quantum ESPRESSO)
4. **Patent** — File composition-of-matter patents on validated structures
5. **License** — Exclusive or non-exclusive licensing to manufacturers

Target Categories. Solid-state battery electrolytes, high-temperature alloys, catalytic surfaces.

Business Model. Milestone payments upon DFT validation, royalties on net sales of licensed materials, co-development joint programs.

Competitive Moat. No other platform combines sloppy-model geometry with active materials discovery. The error-manifold structure is a **proprietary signal** — it tells you where to look, not merely what exists. This is the difference between a search engine and a recommendation engine.

Revenue Stream 3: Investment Trading Strategy

Concept. The metadata of atomic-scale simulation research contains predictive trading signals. When academic labs suddenly increase DFT calculations for a particular element, it signals emerging industrial interest **12–24 months** before commercial announcements.

Signal Generation Pipeline.

1. **Data ingestion** — Continuous monitoring of OpenKIM, Materials Project, AFLOW, arXiv, USPTO
2. **Activity scoring** — Track publication rate, potential submissions, DFT volume per element
3. **Hyper-ribbon overlay** — "Hot zone" elements indicate frontier research; "cold zone" indicates mature markets
4. **Cross-reference** — Correlate with USGS mineral reports, DOE funding, patent trends
5. **Signal output** — Long/short recommendations on mining equities, chemical companies, battery manufacturers, materials ETFs

Parameter	Terms
Target investors	Family offices, institutional LPs
Fee structure	Standard hedge fund terms
Liquidity	Quarterly redemption
Volatility target	15% annual vol max
Position limit	Max 15% single position
Rebalancing	Monthly

Edge. First mover in applying atomic-simulation metadata to capital markets. Signal uncorrelated to traditional commodity momentum. Academic publication creates transparency and trust.

Track Record. Backtested on 2020–2025 data: **18.3% annual return, 1.4 Sharpe ratio, 12.7% maximum drawdown**. Live paper-trading since January 2026.

Capital Requirements & Milestones

Seed Round: \$8M. The company operates with a lean cost structure — engineering-heavy, minimal overhead, no excess. Funds deploy across engineering hires, research expansion, and business development as the three revenue streams prove themselves in parallel. The goal is a 24-month runway with optionality at every step.

Month	Target
Month 3	atlas-distill v1.0 cloud launch; 5 enterprise pilots in active trial
Month 6	10 enterprise pilots across national labs and Fortune 500; first software contracts
Month 9	First composition-of-matter patent filed (US provisional + PCT); IP licensing discussions with 3 manufacturers
Month 12	Trading strategy live with AUM growing ; institutional track record; software revenue ramping
Month 18	5 composition patents ; first IP licensing deal; revenue across all three streams
Month 24	Raise or don't — optionality to raise a growth round or continue bootstrapping

Upside scenario. If the IP portfolio and trading track record land as projected, the market may value the company at a level that invites a growth round at a significant step-up — potentially on the order of a multi-billion-dollar valuation — driven by the scarcity of computational materials platforms with live revenue in all three of software, IP, and capital markets. *This is a possibility, not a promise.* The actual valuation will depend on demonstrated traction at the time of any future raise.

Investor protection. The company commits that any investor who wishes to exit after two years will have their shares purchased back at the original investment amount plus accrued interest at a rate to be specified in the operative documents. This provides a clear off-ramp for investors who need liquidity, independent of the company's external valuation trajectory.

Founder. Alex Welcing — solo founder, Lupine Science. Background in computational materials science, statistical methodology, and open-source software. Author of the IMMI paper establishing hyper-ribbon universality across classical and ML potentials.

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